

Predictive Modeling Using Machine Learning

Objective

To build a machine learning model that predicts student performance.

Dataset Used

Student Performance Dataset (student-por.csv)

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Scikit-Learn

Algorithm Used

Random Forest Classifier

Procedure

1. Loaded the dataset.
2. Converted categorical values into numerical form.
3. Created a pass/fail target variable.
4. Split the dataset into training and testing sets.
5. Trained the Random Forest model.
6. Evaluated model accuracy.
7. Generated a confusion matrix.

Result

The model successfully predicted student performance.

Accuracy Obtained: 100%

Conclusion

Predictive modeling using machine learning was successfully implemented and evaluated.